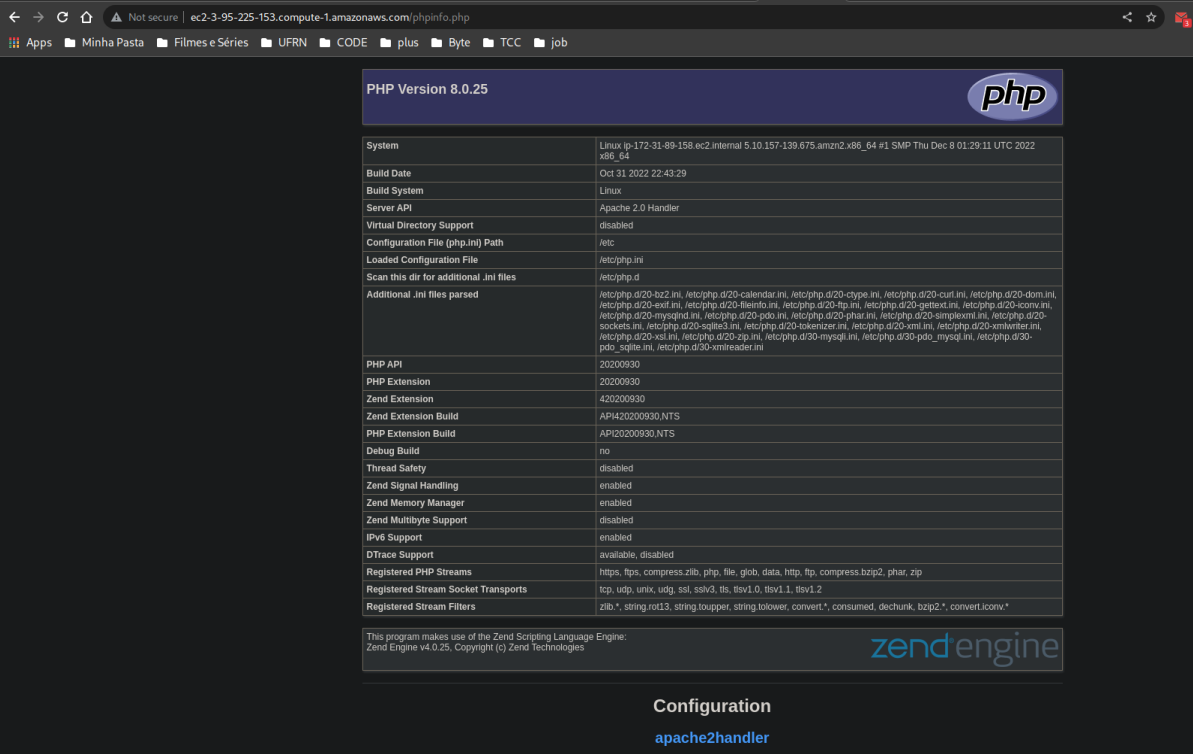


Tarefa 2.1 - Instalação de Serviços da Camada de Aplicação

Reproduzir as instalações dos serviços indicados no Guia de Instalação: Serviços da Camada de Aplicação. Os seguintes serviços devem ser instalados em uma máquina virtual da AWS:

- Servidor WEB Apache
- PHP
- MySQL (MariaDB)
- PHPMyAdmin
- Servidor SMTP Exim
- Servidor de leitura de e-mails POP3

Aluno: Zaú Júlio Araújo Galvão



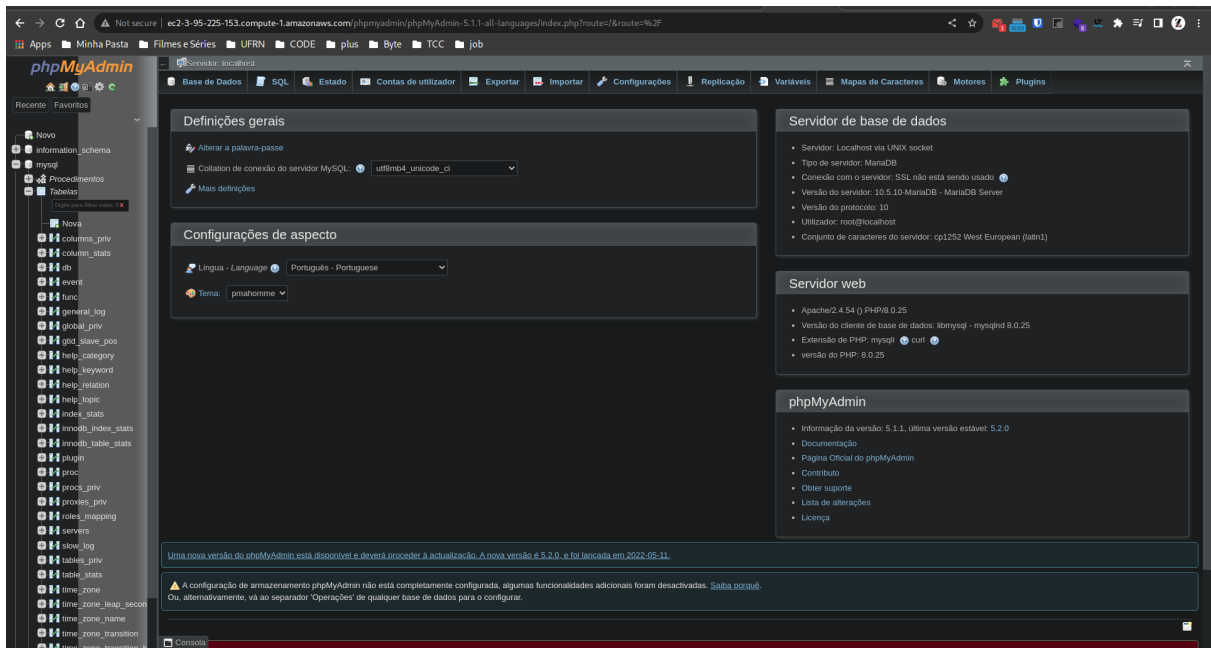
The screenshot shows the PHP Info page for PHP Version 8.0.25. The page is displayed in a web browser with the address bar showing the URL `ec2-3-95-225-153.compute-1.amazonaws.com/phpinfo.php`. The page features a dark blue header with the PHP logo and a table of system and configuration details.

PHP Version 8.0.25	
System	Linux ip-172-31-89-158.ec2.internal 5.10.157-139.675.amzn2.x86_64 #1 SMP Thu Dec 8 01:29:11 UTC 2022 x86_64
Build Date	Oct 31 2022 22:43:29
Build System	Linux
Server API	Apache 2.0 Handler
Virtual Directory Support	disabled
Configuration File (php.ini) Path	/etc
Loaded Configuration File	/etc/php.ini
Scan this dir for additional .ini files	/etc/php.d
Additional .ini files parsed	/etc/php.d/20-bz2.ini, /etc/php.d/20-calendar.ini, /etc/php.d/20-ctype.ini, /etc/php.d/20-curl.ini, /etc/php.d/20-dom.ini, /etc/php.d/20-exif.ini, /etc/php.d/20-fileinfo.ini, /etc/php.d/20-ftp.ini, /etc/php.d/20-gettext.ini, /etc/php.d/20-iconv.ini, /etc/php.d/20-mysqld.ini, /etc/php.d/20-pdo.ini, /etc/php.d/20-phar.ini, /etc/php.d/20-simplexml.ini, /etc/php.d/20-sockets.ini, /etc/php.d/20-sysvmsg.ini, /etc/php.d/20-tokenizer.ini, /etc/php.d/20-xml.ini, /etc/php.d/20-xmlreader.ini, /etc/php.d/20-xmlwriter.ini, /etc/php.d/20-xsl.ini, /etc/php.d/20-zip.ini, /etc/php.d/30-mysql.ini, /etc/php.d/30-pdo_mysql.ini, /etc/php.d/30-pdo_sqlite.ini, /etc/php.d/30-xmireader.ini
PHP API	20200930
PHP Extension	20200930
Zend Extension	420200930
Zend Extension Build	API420200930.NTS
PHP Extension Build	API20200930.NTS
Debug Build	no
Thread Safety	disabled
Zend Signal Handling	enabled
Zend Memory Manager	enabled
Zend Multibyte Support	disabled
IPv6 Support	enabled
DTrace Support	available, disabled
Registered PHP Streams	https, ftps, compress.zlib, php, file, glob, data, http, ftp, compress.bzip2, phar, zip
Registered Stream Socket Transports	tcp, udp, unix, udg, ssl, sshv3, tls, tlsv1.0, tlsv1.1, tlsv1.2
Registered Stream Filters	zlib.*, string.rot13, string.toupper, string.tolower, convert.*, consumed, dechunk, bzip2.*, convert.iconv.*

This program makes use of the Zend Scripting Language Engine:
Zend Engine v4.0.25, Copyright (c) Zend Technologies

zendengine

Configuration
apache2handler



```
ec2-user@ip-172-31-89-158 /var/www/html/phpmyadmin
$ exim --version

Exim version 4.96 #2 built 22-Nov-2022 14:42:46
Copyright (c) University of Cambridge, 1995 - 2018
(c) The Exim Maintainers and contributors in ACKNOWLEDGMENTS file, 2007 - 2022
Berkeley DB: Berkeley DB 5.3.21: (May 11, 2012)
Support for: crypteq iconv() IPv6 PAM Perl Expand_difunc OpenSSL TLS_resume Content_Scanning DANE DKIM D
MARC DNSSEC Event OSCP PIPECONNECT PRDR PROXY Queue_Ramp SOCKS SPF TCP_Fast_Open
Lookups (built-in): lsearch wildlsearch nwildlsearch iplsearch cdb dbm dbm3jz dbmnz dnsdb dsearch ldap ld
apdn ldapm nis nis0 nisplus passwd sqlite
Authenticators: cram_md5 cyrus_sasl dovecot gssapi plaintext spa tls
Routers: accept dnslookup ipliteral manualroute queryprogram redirect
Transports: appendfile/maildir/mailstore/mbx autoreply lmtp pipe smtp
Malware: f-protfd f-prot6d drweb fsecure sophie clamd avast sock cmdline
Fixed never_users: 0
Configure owner: 0:0
Size of off_t: 8
Configuration file is /etc/exim/exim.conf
ec2-user@ip-172-31-89-158 /var/www/html/phpmyadmin
$
```

```
ec2-user@ip-172-31-89-158 ~$ sudo systemctl status httpd
● httpd.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/httpd.service; enabled; vendor preset: disabled)
   Drop-In: /usr/lib/systemd/system/httpd.service.d
           └─php-fpm.conf
   Active: active (running) since Mon 2022-12-19 02:31:23 UTC; 12min ago
     Docs: man:httpd.service(8)
   Main PID: 5838 (httpd)
   Status: "Total requests: 287; Idle/Busy workers 100/0;Requests/sec: 0.389; Bytes served/sec: 3.2KB/se

CGroup: /system.slice/httpd.service
├─5838 /usr/sbin/httpd -DFOREGROUND
├─5840 /usr/sbin/httpd -DFOREGROUND
├─6060 /usr/sbin/httpd -DFOREGROUND
├─6072 /usr/sbin/httpd -DFOREGROUND
├─6073 /usr/sbin/httpd -DFOREGROUND
├─6084 /usr/sbin/httpd -DFOREGROUND
├─6085 /usr/sbin/httpd -DFOREGROUND
├─6086 /usr/sbin/httpd -DFOREGROUND
├─6087 /usr/sbin/httpd -DFOREGROUND
├─6110 /usr/sbin/httpd -DFOREGROUND
└─6296 /usr/sbin/httpd -DFOREGROUND
```

Apacch2

```
ec2-user@ip-172-31-89-158 ~$ sudo systemctl status exim.service
● exim.service - Exim Mail Transport Agent
   Loaded: loaded (/usr/lib/systemd/system/exim.service; enabled; vendor preset: disabled)
   Active: inactive (dead) since Mon 2022-12-19 02:45:42 UTC; 32s ago
   Process: 7468 ExecStart=/usr/sbin/exim -bd -q${QUEUE} (code=exited, status=0/SUCCESS)
   Main PID: 7468 (code=exited, status=0/SUCCESS)

Dec 19 02:44:53 ip-172-31-89-158.ec2.internal systemd[1]: Starting Exim Mail Transport Agent...
Dec 19 02:44:53 ip-172-31-89-158.ec2.internal exim-gm-cert[7452]: Generating exim certificate: success
Dec 19 02:44:53 ip-172-31-89-158.ec2.internal systemd[1]: Started Exim Mail Transport Agent.
Dec 19 02:45:42 ip-172-31-89-158.ec2.internal systemd[1]: Stopping Exim Mail Transport Agent...
Dec 19 02:45:42 ip-172-31-89-158.ec2.internal systemd[1]: Stopped Exim Mail Transport Agent.

Hint: Some lines were ellipsized, use -l to show in full.
```

Email

```
ec2-user@ip-172-31-89-158 ~$ sudo systemctl status mariadb
● mariadb.service - MariaDB 10.5 database server
   Loaded: loaded (/usr/lib/systemd/system/mariadb.service; disabled; vendor preset: disabled)
   Active: active (running) since Mon 2022-12-19 02:31:26 UTC; 12min ago
     Docs: man:mariadb(8)
           https://mariadb.com/kb/en/library/systemd/
   Main PID: 5975 (mariadbd)
   Status: "Taking your SQL requests now..."
   CGroup: /system.slice/mariadb.service
           └─5975 /usr/libexec/mariadbd --basedir=/usr

Dec 19 02:31:26 ip-172-31-89-158.ec2.internal systemd[1]: Stopped MariaDB 10.5 database server.
Dec 19 02:31:26 ip-172-31-89-158.ec2.internal systemd[1]: Starting MariaDB 10.5 database server...
Dec 19 02:31:26 ip-172-31-89-158.ec2.internal mariadb-prepare-db-dir[5936]: Database MariaDB is proba...
Dec 19 02:31:26 ip-172-31-89-158.ec2.internal mariadbd[5975]: 2022-12-19 2:31:26 0 [Note] /usr/lib....
Dec 19 02:31:26 ip-172-31-89-158.ec2.internal systemd[1]: Started MariaDB 10.5 database server.

Hint: Some lines were ellipsized, use -l to show in full.
```

Mariadb

```
ec2-user@ip-172-31-89-158 ~$ sudo systemctl status postfix.service
● postfix.service - Postfix Mail Transport Agent
   Loaded: loaded (/usr/lib/systemd/system/postfix.service; enabled; vendor preset: disabled)
   Active: active (running) since Mon 2022-12-19 02:45:43 UTC; 2s ago
   Process: 7572 ExecStart=/usr/sbin/postfix start (code=exited, status=0/SUCCESS)
   Process: 7560 ExecStartPre=/usr/libexec/postfix/chroot-update (code=exited, status=0/SUCCESS)
   Process: 7563 ExecStartPre=/usr/libexec/postfix/aliasesdb (code=exited, status=0/SUCCESS)
   Main PID: 7645 (master)
   CGroup: /system.slice/postfix.service
           └─7645 /usr/libexec/postfix/master -w
             └─7647 pickup -l -t unix -u
               └─7648 qmgr -l -t unix -u

Dec 19 02:45:42 ip-172-31-89-158.ec2.internal systemd[1]: Starting Postfix Mail Transport Agent...
Dec 19 02:45:43 ip-172-31-89-158.ec2.internal postfix/postfix-script[7643]: starting the Postfix mail...
Dec 19 02:45:43 ip-172-31-89-158.ec2.internal postfix/master[7645]: daemon started -- version 2.10.1...x
Dec 19 02:45:43 ip-172-31-89-158.ec2.internal systemd[1]: Started Postfix Mail Transport Agent.

Hint: Some lines were ellipsized, use -l to show in full.
```

Pop3